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Saveetha Institute of Medical And Technical Sciences
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HYBRID VS. MULTI-CLOUD ARCHITECTURE

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**Course : Cloud Computing and
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Code : CSA 1506

Modern Hybrid Cloud Paradigm

-  **Private / On-Premise Infrastructure:** Secure enterprise data center holding confidential records, financial data, and customer for regulatory compliance and low latency.
-  **Public Cloud Scale:** AWS, Azure, and GCP provide elastic computing power on demand, drastically lowering infrastructure costs through pay-as-you-go models.
-  **Distributed Edge Computing:** Processes data closer to users and IoT devices, minimizing backhaul latency and maximizing real-time performance.



Hybrid vs. Multi-Cloud



Hybrid Cloud

Combines a company's private infrastructure (on-premise or private cloud) with public cloud services.

Keeps sensitive information highly secure while leveraging public cloud burst capacity when computing power spikes.



Multi-Cloud

Utilizes services from multiple public cloud providers (e.g., AWS, Microsoft Azure, Google Cloud Platform).

Prevents vendor lock-in and allows enterprises to pick best-of-breed specialized tools from each vendor.



Data Gravity

As enterprise data grows, moving large amounts of data becomes cost-prohibitive and time-consuming.

Organizations deploy applications closer to data sources via distributed edge computing rather than shifting bulk data.

Unified Connectivity & Management



Unified Connectivity

Dedicated high-speed network pipes establish secure communication between private data centers and public clouds.

- **AWS Direct Connect:** Encrypted, low-latency private link.
- **Azure ExpressRoute:** High-throughput private circuit.



Centralized Management

Operating multi-vendor platforms requires a single pane of glass for deployment, security, and health monitoring.

- **Kubernetes:** Container orchestration engine.
- **Google Anthos & Azure Arc:** Multi-cloud control plane.

Interconnection & Orchestration



Cloud Bursting

Run baseline workloads privately; seamlessly burst spikes into public cloud capacity without hardware investment.



Governance

Adhere strictly to GDPR, HIPAA, and data sovereignty laws by keeping local while processing non-sensitive workloads on public cloud.



CAPEX vs. OPEX

Shift from heavy capital expenditure in physical servers to predictable operational expenditure via elasticity.

Strategic Enterprise Drivers



Business Continuity

Multi-provider redundancy ensures zero operational downtime. If one cloud provider suffers an outage, failover workloads automatically execute across an alternate cloud.



Enhanced Security Posture

Isolate key intellectual property and sensitive customer data inside protected private infrastructure, isolating risks while consuming public compute safely.



Cost Optimization

Dynamically select competitive pricing models and discount tiers across multiple vendors for distinct computational or storage requirements.



Global Expansion

Deploy application nodes internationally within local cloud regions, reducing latency for localized users and meeting local regulatory rules.

Industry Landscape & Adoption

80%+

Global Enterprise Adoption

The Standard for Modern Infrastructure

Over 80 percent of global enterprises have actively transitioned to Hybrid or Multi-Cloud operational models as their core IT strategy.



Banking & Healthcare: Strict compliance, high availability, and secure transactions.



Manufacturing & Retail: Edge analytics and real-time inventory responsiveness.



Telecommunications: Massively distributed edge networks with multi-cloud backends.

Core Architectural Challenges



Operational Complexity

Managing heterogeneous cloud platforms requires multi-skilled talent due to varied native tools, APIs, and administrative interfaces.



Security & Identity

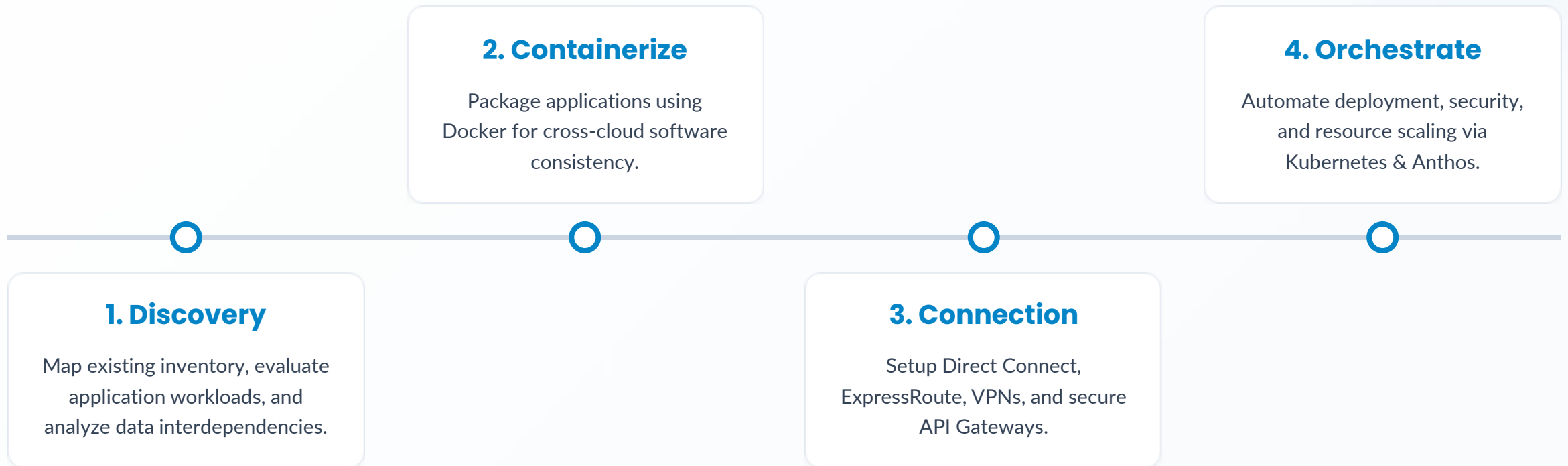
Maintaining uniform IAM policies, role-based access control, encryption standards, and threat auditing across multiple clouds is difficult.



FinOps & Egress Costs

Moving substantial datasets across vendor cloud boundaries generates unexpected data egress fees, demanding strict FinOps governance.

Enterprise Migration Roadmap





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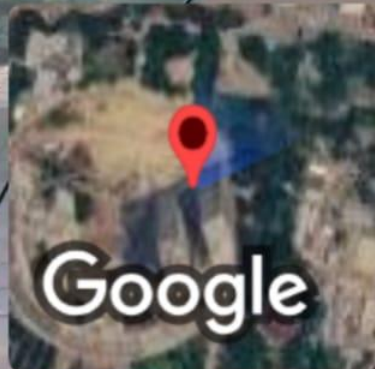
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